

EcoPoints: A Smart Waste Collection and Reward Redemption System

Waste mismanagement poses a critical global challenge, exacerbating environmental pollution, public health risks, and inefficient recycling due to low public engagement and lack of incentives for proper segregation and disposal. This capstone project introduces *EcoPoints*, an innovative IoT-based system designed to foster sustainable waste practices through technology and gamification. Utilizing ESP32 microcontrollers integrated with sensors (e.g., ESP32 camera module for enhanced functionality), the system automates real-time waste tracking, capturing data on disposal type and quantity, which is transmitted to a user-friendly web dashboard for monitoring and analysis.

The primary objective is to develop a reward-driven platform that incentivizes responsible behavior by awarding EcoPoints redeemable for tangible incentives through points redemption, and every 100 (one hundred) points is equivalent to 1 (one) peso. And then the redemption of your converted points, it will serve you for a free printing (as payment) at SOICT office, thereby enhancing efficiency, transparency, and community participation.

By evaluating current practices and integrating smart sensors with a reward model, EcoPoints automates waste detection and quantification to assign points dynamically (e.g., 10 points per piece of recyclables), enabling users to view personalized progress reports, leaderboards for gamified competition, and educational modules on segregation benefits via the dashboard. This targeted approach not only streamlines data processing to minimize errors but also supports scalability through modular hardware and cloud-based storage.